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Research paper

Artificial intelligence based-improving reservoir management: An Attention-Guided Fusion Model for predicting injector–producer connectivity

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ABSTRACT

The existing oil reservoir demonstrated suboptimal inter-well connectivity, leading to irregular depletion and reduced overall production efficiency. This article demonstrates the Attention-Guided Fusion Model for Injector–Producer Connectivity Estimation (AGFM). The model has an attention mechanism in the first path, pulling discernment from the relationships between injectors and producers through the training phase, extracting the attention weight. This attention weight is then devoted to the second path, utilising a Long-Short-Term Memory (LSTM)-based architecture. The first path is only to the training stage. In contrast, the second path is used during training and testing, improving the ability of the model to find a more significant representation of the data. This makes the model robust enough to predict reservoir performance and interconnectivity, giving valuable insights to optimise field operations. The AGFM undergoes an evaluation with two different injection liquids (carbon dioxide (CO_2) and water) in three scenarios: all water and all CO2 alternating between water and CO2 as a flooding liquid. The evaluation emphasises the efficacy of the model in all scenarios, making it a practical tool for estimating reservoir connectivity and enhancing oil recovery strategies. The water alternating gas (WAG) process performed high accuracy rates, with 82.1% for oil production, 86.8% for water production, and 86.9% for gas production. Our proposed method consistently demonstrates superior performance through comprehensive experimentation and rigorous analysis compared to existing approaches. The results reveal spatial interwell connectivity, confirming the efficacy and potential of our method as a more effective solution for reservoir recovery.

1. Introduction

Injector–producer connectivity analysis decides the degree of association between the injector and the producer wells in an oil or gas reservoir. This information is essential for reservoir management and field growth, as it can be used to optimise production speed and water flood or carbon dioxide (CO2) flood strategies (Kashkooli et al., 2022).

Injector–producer connectivity appears from the point of view that reservoirs are often heterogeneous, indicating that the permeability and porosity of the stone can change significantly from one place to another (Al-Mudhafar et al., 2023). This can make it challenging to forecast the flow of fluids between the injection and producer wells. Knowing the optimal injection rate of water or CO2 increases the recovery of reservoir oil and gas. In addition, determining the locations of the infill wells and their optimal spacing is vital to optimal reservoir management. This is because a wide, healthy spacing can lead to some of the hydrocarbon-bearing sands needing to be swept efficiently. In contrast, close spacing can cause interference between wells and reduce the volumes of fluid drained by wells producing (Annan Boah et al., 2019). Thus, careful assessment of inter-well connectivity is a crucial procedure for comprehending communication among multiple producers and injectors. Therefore, enhanced oil recovery (EOR) processes use a reservoir management technique. In the injection of water-alternating gas (WAG), water and gas are alternately injected into an oil reservoir

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to improve oil recovery by optimising sweep efficiency.

Different artificial intelligence (AI)-based methods are proposed for this task. Optimisation techniques have been used to find the best possible injector–producer connectivity for wells in a petroleum reservoir. Gradient-free optimisation algorithms are often used for this purpose (Akkem et al., 2023b).

One of the most popular gradient-free algorithms is particle swarm optimisation (PSO) (Du et al., 2023). However, due to the number of particles, the inertia weight, and the acceleration coefficients, PSO can be sensitive to the choice of these parameters and may not always converge to the global optimum solution. Genetic algorithms (GA) are stochastic optimisation algorithms that are used to optimise injector–producer connectivity in oil reservoirs (Zakaria et al., 2011). GAs require many evaluations of the objective function, which leads to an increase in their computations and has no guarantee to converge to the global optimum solution.

To avoid sticking in local optima, artificial neural networks (ANN) are also used to identify injector–producer relationships (Saikia et al., 2020; Nwachukwu et al., 2022; Zhou et al., 2021; Xiong and Lee, 2020; Demiryurek et al., 2008). However, ANN models can be limited in their ability to model complex systems with many variables. Furthermore, ANN models are prone to overfitting, which means they can accurately predict the production performance of reservoirs in the training dataset. Still, they may not be able to generalise to new reservoirs.

ANNs may be unable to model complex systems that involve many factors influencing the reservoir's production performance. Thus, deep learning approaches for the estimation of injector–producer connectivity have been proposed. These approaches include a convolutional neural network (CNN) to learn the relationship between injection rate, production rate, and seismic data (Jiang et al., 2023c; Du et al., 2020; Carmichael and Liao, 2022). CNNs are local operations, meaning they only consider each point's immediate neighbours when making predictions. This makes them less effective at finding long-range dependencies, such as the relationship between the injection well and the producer that is far apart.

Due to CNN's local operation, it cannot consider relationships between data points (i.e., wells) that are far apart; recurrent Neural Networks (RNNs) have been used to find the connectivity of the injector and producer wells (Jiang et al., 2023a; Sen et al., 2022b). RNNs could adequately find long-range dependencies through long-range time steps but ignore each injector's immediate relationships to all producer wells.

In a reservoir, numerous producer wells might be immediately affected by an injector well simultaneously. By composition, RNNs process data sequentially, concentrating on how past states impact the current state rather than simultaneously considering direct spatial exchanges between wells that occur at the same time step. Ensemble learning (Wang et al., 2017) allows the ensemble to capture spatial and temporal dependencies simultaneously, providing a complete view of the exchanges between wells. However, ensemble methods are difficult to decode, making it challenging to comprehend individual exchanges and the underlying physical relationships in the reservoir. To avoid this, Graph Neural Networks (GNNs) (Huang et al., 2024; Saihood et al., 2024) consider each well (injector or producer) as a node, and the connections between them are edges. GNNs discover the spatial relationships between nodes and catch local and global dependencies by handing transmissions along these edges. However, GNNs may lose their ability to effectively capture the entire system's dynamics when the graph structure's scalability changes or becomes too large.

To find inter-well connectivity in arbitrarily distributed wells in specific reservoirs, this article proposed an Attention-guided Fusion Model (AGFM). AGFM consists of two paths; the first consists of computing the relationships between each injector well and all producer wells through three-dimensional matrices called Injectors-to-Producer Maps (IPMs). IPMs consist of three 2D matrices representing the relationships between injectors and producers fused through three channels of produced gas, oil, and water in the producer wells, and IPM model guidance to find attention weights through a deep neural network-based architecture that can be applied to the second path. The second path consists of an LSTM-based architecture to predict the liquids produced based on applied attention weights obtained from the first path, highlighting the essential injectors for producer relations. The first path is used only through the training phase and the second is used in the training and testing phases.

Unlike conventional approaches that tend to subtle spatial and temporal relationships between injectors and producers, AGFM presents a two-path architecture that leverages an attention mechanism to determine effective exchanges between wells during the training phase. This attention weight is then involved in an LSTM-based architecture to model long-term dependencies and forecast reservoir production more accurately. By integrating the powers of attention and LSTM-based architectures, the AGFM delivers a complete interpretation of inter-well dynamics and reaches higher prediction accuracy, as shown by its powerful performance across different injection scenarios.

The issues addressed in this article are (1) most of the methods state-of-the-art used to train and test their suggested method using single flooding of water or CO2; in this article, we used multiple scenarios using water, CO2 or alternating water and CO2 injections to satisfy the WAG property. (2) Due to the spatial connectivity between injectors and producers and the complex connections between them, injecting water or CO2 is not necessary to produce the same amount of oil, gas or water, making the training process challenging to find all these relationships due to the lack of available data sets. The proposed model uses attention-based fusion in a dual-path scenario to guide the model to focus on essential regions that can find accurate connectivity among injectors and producers. (3) The injector-to-producer connection is a global relationship; using a meta-heuristic algorithm can be stuck in local optima. The proposed method can find attentive global dependencies based on the IPM guidance mechanism.

The main contributions of this article are:

1. We propose a novel AGFM for injector–producer connectivity, using an attention mechanism to allocate weights to vital connectivity aspects, improving the model's learning for improved data representations.
2. Propose an IPM-based guidance method to guide the model in identifying accurate injector-to-producer relationships.
3. We evaluated the proposed model using multiple injection liquids (CO2 and water) and alternative water and CO2 injection scenarios. This evaluation demonstrates the effectiveness of the proposed model for various injection liquids and ensures WAG property.

The rest of this article is structured as follows: in Section 2, we introduce related works on inter-well connectivity methods used recently, and in Section 3 explain the proposed AGFM model with the IPM guidance used in the model. In Section 4, we present our experiments and their results in different scenarios and compare them with the state-of-the-art. Finally, Section 5 concludes the paper and provides an outlook for future work.

2. Related studies

Inter-well connectivity, often called injector–producer connectivity, has been a subject of significant interest and research in reservoir engineering and hydrocarbon recovery. This section reviews vital work that has contributed to the interpretation and analysis of these intricate subsurface fluid interactions.

Intelligent farming uses machine learning and deep learning techniques for crop yield prediction, soil analysis, and water management tasks. They emphasise their role in optimising agricultural practices using support vector machines, decision trees, ensemble learning, and recurrent neural networks. Techniques such as Extreme Learning Machines (ELM) and Radial Basis Function Networks (RBFN) are also

Table 1
Issues and datasets of the methods in state-of-the-art.

Methods	Dataset	Issues
PSO (Du et al., 2023)	Carbonate reservoir in the Middle East	Due to the number of particles, the inertia weight, and the acceleration coefficients, PSO can be sensitive to the choice of these parameters and may not always converge to the global optimum solution.
GA (Zakaria et al., 2011)	West Africa offshore field case	GAs require many evaluations of the objective function, which leads to an increase in their computations and has no guarantee to converge to the global optimum solution.
ANN (Xiong and Lee, 2020)	Non-real dataset	The use of the TOUGH2 code to generate training data in the article represents a deficiency, as the derived nonreal dataset may need to serve as substantial evidence for effectively training the models.
RNNs (Jiang et al., 2023a)	Non-explained dataset	RNNs could adequately find long-range dependencies through long-range time steps but ignore each injector's immediate relationships to all producer wells.
LSTM-EFAST (Du et al., 2020)	Non-real dataset	The LSTM-EFAST method assumes that the input factors are independent and uniformly distributed in the input space. This may not be true for some real-world scenarios where the input factors may have correlations or non-uniform distributions. This may affect the quality of the sensitivity analysis and the interpretation of the results.
Ensemble learning methods (Wang et al., 2017)	North of China oil field	The StoSAG algorithm may suffer from slow convergence and poor performance when dealing with complex optimisation problems with high dimensions, multiple local optima and non-linear constraints.
Random Forests (Ma et al., 2023)	South China Sea area	Wrong tuning of the random forest parameters leads to wrong results.
Statistical Recurrent Unit (SRU) (Chen et al., 2022)	Two types of data: one is the output of simulation (synthetic data), and the second is field-scale CO2 EOR projects (no more details mentioned)	In their method, the authors did not consider the reservoir's underlying changes due to pressure depletion, fluid movement, etc., over time.
Graph Neural Network (GNN) (Huang et al., 2024)	Shengli Oilfield.	Requires large amounts of high-quality production and geological data; limited scalability in highly heterogeneous reservoirs.
Physical Knowledge Fusion Neural Network (PKFNN) (Jiang et al., 2023b)	field data from waterflooding experiments.	Model complexity increases training time; difficult to generalise to other field scenarios without extensive retraining.

discussed in Akkem et al. (2023a, 2024).

Optimisation of well placement could be handled through metaheuristic algorithms (Onuh et al., 2017; Yuan et al., 2021). Onuh et al. (2017) used a genetic search heuristic algorithm to intricately build well locations within the theoretical Finch reservoir using a black oil simulator. The overarching objective is to establish optimal production results in amidst varying scenarios, and to this end, eight distinct configurations of good placements were meticulously prepared. Yuan et al. (2021) use a methodology that leverages the power of particle swarm optimisation (PSO) alongside the predictive abilities of CatBoost, presenting a comprehensive framework for the analysis of inter-well connectivity.

In reservoir engineering, the integration of machine learning techniques has appeared as a transformative approach to solving the complexities of inter-well connectivity (Paul et al., 2021; Framework, 2021). Moreover, deep learning architectures (Cheng et al., 2023) have been used to model complex reservoir dynamics. Neural networks, particularly recurrent neural networks (RNNs) (Sen et al., 2022b; Cheng et al., 2020) and extended short-term memory networks (LSTMs) (Du et al., 2020) can catch temporal dependencies within inter-well connectivity datasets. By training on recorded reservoir behaviour, these models contribute to accurate predictions, helping to optimise good placements for improved hydrocarbon recovery. Ensemble learning methods (Wang et al., 2017), such as Random Forests (Ma et al., 2023; Yan et al., 2021) and gradient boost (Ramalho et al., 2024; Wang et al., 2021), have shown the ability to enhance predictive accuracy for inter-well connectivity. Connecting multiple models' strengths, ensemble methods mitigate overfitting and contribute to more reliable predictions, mainly when dealing with heterogeneous reservoir conditions.

In recent years, ANNs have shown significant opportunities to capture complex connections within reservoir datasets, making them critical participants in understanding fluid dynamics between wells (Hamidi and Rafati, 2012). Illuminated by the neural construction of the human brain, these computational models excel in discovering complex patterns and dependencies inherent in the reservoir data. The adaptability of ANNs enables them to analyse and predict inter-well connectivity by processing information associated with reservoir attributes, fluid behaviour, and well relations (Hernandez-Ambato et al., 2017).

The various methods studied in the related works section emphasise the multifaceted nature of inter-well connectivity research. The reservoir engineering topography has detected innovative techniques, from metaheuristic algorithms that optimise well placements to the transformative integration of machine learning and deep learning architectures. Ensemble learning methods further contribute to the robustness of predictions and handle the challenges posed by heterogeneous reservoir requirements. The increasing importance of Artificial Neural Networks signifies a paradigm shift, presenting significant possibilities for capturing the intricate dynamics within reservoir datasets. As reservoir engineers seek optimal hydrocarbon recovery through strategic well placements, the amalgamation of these approaches promises to chart new boundaries in reservoir management and contribute to the sustainable exploitation of hydrocarbon resources.

Table 1 comprehensively presents several optimisation techniques for well placement and management problems in oil reservoir control. When used in a carbonate reservoir in the Middle East, the PSO is noted for its sensitivity to parameter options, such as the number of particles, inertia weight and acceleration coefficients, which could lead to convergence problems. Genetic algorithms (GA) used in a West African offshore field case are criticised for the high number of objective function evaluations, resulting in increased computational requirements and no guarantee of convergence to the global optimum solution.

ANN, using a nonreal dataset generated by the TOUGH2 code, faces challenges because the training data is required to be more vital in effectively defining real-world scenarios.RNNs, dealing with an unexplained dataset, demonstrate strengths in capturing long-range dependencies but may overlook direct connections between injectors

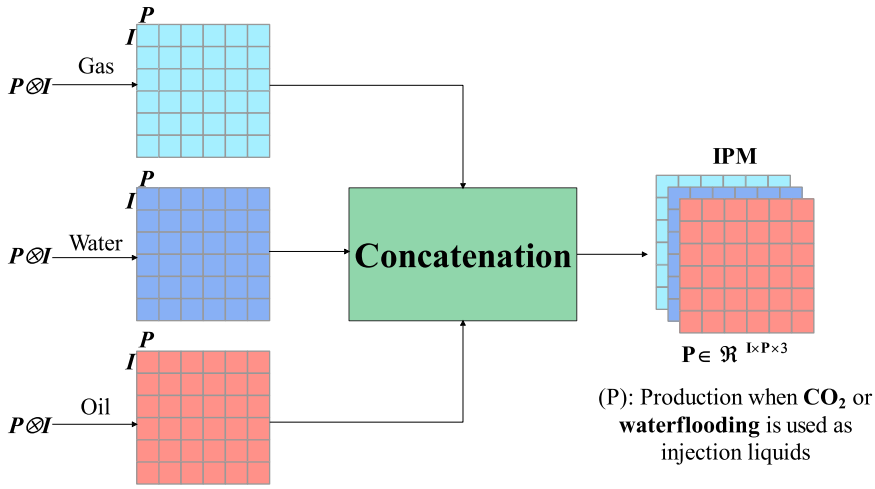


Fig. 1. IPM computations process.

and producer wells.LSTM-EFAST employs a nonreal dataset, considering independent and uniformly distributed input factors, potentially impacting sensitivity analysis quality in scenarios with correlated or nonuniformly distributed input factors.

Ensemble learning methods, specifically the StoSAG algorithm, applied to a North China oil field, face potential issues with slow convergence and poor performance in complex optimisation problems characterised by high dimensions, multiple local optima, and nonlinear constraints. Random forests can produce accurate results in the South China Sea area when properly tuned. The statistical repeat unit (SRU), using two data types, including simulation output and field-scale CO2 EOR projects, does not account for the underlying changes in the reservoir over time, such as pressure depletion and fluid movement.

Optimisation algorithms, such as those used in machine learning and numerical optimisation tasks, exhibit specific challenges and limitations. First, they may require many iterations to achieve convergence, mainly when dealing with high-dimensional or ill-conditioned problem spaces. Additionally, these algorithms might encounter difficulties navigating non-convex functions, often getting trapped in local minima or saddle points. The sensitivity to the choice of step size or learning rate is another concern which influences both the convergence speed and stability of these algorithms. Similarly, its efficacy may be compromised when faced with constraints, noise, or uncertainty in the objective function or the input data. Lastly, these algorithms can incur high communication costs in distributed computing scenarios that involve multiple workers, as they require the transmission of the full gradient vector at each iteration, posing a potential scalability challenge (Sen et al., 2022a; Zhao et al., 2024).

3. Method

In this study, we proposed AGFM to assess interwell connectivity within reservoirs in the petroleum field. In the same context, we introduced IPM computations, in which we consider the relationships among the injection and producer wells. Next, we explained the first path of the proposed model, in which a 3D-CNN-based attention model is used to extract attention weights that are applied to parallel the second path to enhancing the significance and prioritising specific regions based on the associations among actual output data in influencing the model's focus on particular areas. In addition, we explain the second path of the proposed model in which an LSTM-based architecture is used to predict the output liquids of the producers.

3.1. Dataset

Our dataset consisted of a field containing nine injectors and 20 producer wells in the Denver City Field in West Texas. Based on the type of fluids injected into the injector well, the producer wells produce an amount of oil, gas, and water. The dataset records two types of injection fluids: CO2 and water. Missing values are controlled to guarantee data completeness by dropping rows, including empty values, resulting in 184 steps of production of oil, gas, and water recorded monthly from the reservoir once in terms of CO2 and Water as injector fluid once again. We divide 80% of the data used for training and 20% for testing.

3.2. Proposed AGFM

This article examines the efficacy of the proposed method by analysing two distinct scenarios. We split the time steps (that is, the monthly recorded states) in the initial scenario when CO2 is used as an injection fluid. Let I_c refer to the injector wells when CO2 is employed as an injection fluid, P^G, P^W, and P^O refer to the gas, water, and oil produced in the producer wells.

The IPM guide captures the spatial associations between injectors and producers utilising three-dimensional matrices that emphasise key production metrics (that is, gas, oil, and water). These IPMs yield attention weights, guiding the model to concentrate on influential inter-well interchanges. The LSTM is then utilised to capture temporal dependencies, as it excels at handling sequential data and maintaining long-term information. IPM guidance and LSTM allow the model to forecast production movements and inter-well connectivity accurately, improving general prediction performance.

3.3. IPM computations

To find the effect of the injection rate on the production rate in terms of CO2, we compute the relationships based on the equations below. (1)–(3).

$$I_c \otimes P^G = I_g, \quad I_g \in \mathbb{R}^{I\times P} \tag{1}$$

$$I_c \otimes P^W = I_w, \quad I_w \in \mathbb{R}^{I\times W} \tag{2}$$

$$I_c \otimes P^O = I_o, \quad I_o \in \mathbb{R}^{I\times O} \tag{3}$$

where I_g, I_w and I_o refer to the IPMs of CO2 and the gas, water, and oil produced calculated using matrix multiplication between injection and production rate. We fused the computed matrices using the concatenation results in one 3D matrix ($P_c \in R^{I\times P\times 3}$) to obtain the input

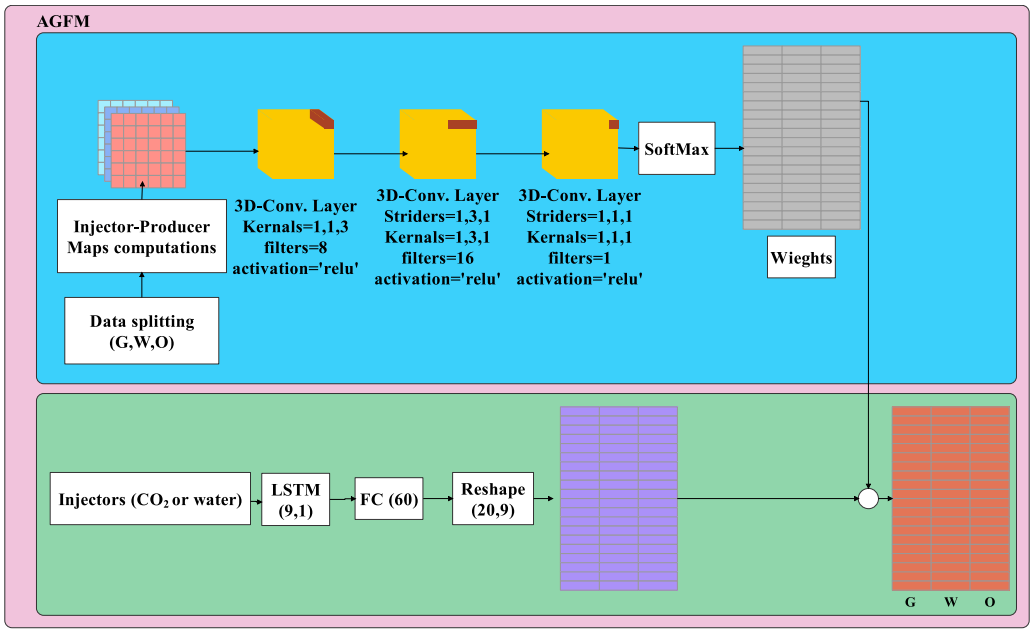


Fig. 2. AGFM model architecture.

matrices for our proposed model. (Fig. 1).

In the same context, we compute the injector–producer relationship considering water as an injection fluid. (4)–(6).

$$I_w \otimes P^G = I_g, \quad I_g \in \mathbb{R}^{I \times P} \tag{4}$$

$$I_w \otimes P^W = I_w, \quad I_w \in \mathbb{R}^{I \times W} \tag{5}$$

$$I_w \otimes P^O = I_o, \quad I_o \in \mathbb{R}^{I \times O} \tag{6}$$

concatenating the computed matrices results in $P_w \in R^{I \times P \times 3}$. In addition, the P_c and P_w computed for 184 time steps will be used as input to train and assess the proposed model. Computing the IPMs can reveal the deep relationships between the producers and injectors and how feeding Water or CO2 to the injectors can affect the producers' production.

3.4. Proposed IPM guidance model

The computed IPMs feed into a deep guidance model to capture relationships among injectors and producers through an adequate three-dimensional convolutional layer. These relationships are calculated as 2D matrices stacked for gas, water, and oil to obtain 3D matrices for each time step. This section explains the proposed guidance for the IPM model (IPM-MG). 3D-conv operations are applied in the first layer with a size kernel (1,1,3). The parameters are set to characterise the spatial relationships between injected CO2 and the gas, water, and oil produced, which are then represented in 2D maps. Next, another 3D-conv operation is applied with a kennel size of (1,3,1) and striders of (1,3,1). This setting enables the model to associate the nine injectors with three columns that represent the gas, water, and oil produced for the P wells. Another 3D-conv layer is added with a kernel size of (1,1,1) and striders of (1,1,1). This layer functions as an embedding layer that projects the output of the last layer into a space p by 3, where p is the number of wells and 3 is the number of liquids produced (i.e. oil, water, and gas). Then, the $p \times 3$ attention scores produced are applied to the SoftMax function to obtain the attention weights. Then, these attention weights are applied through element-wise multiplication to the last layer of the LSTM-based model architecture in the second path. The output of the AGFM model is the oil, gas and water of all producers when water or CO2 is in the injector wells. We use the IPM guidance model only in the training phase, and in the testing phase, we employ only the LSTM-based model architecture path.

3.5. Proposed IPM guidance-based attentive recurrent model (IPM-GAR)

The second trajectory within the proposed model is delineated by providing input injection data to predict producers' production rates. This trajectory encompasses three stratified layers, beginning with a Long-Short-Term Memory (LSTM) layer. The LSTM layer assimilates the input of the injector wells, specifically the introduction of water or CO2 into the wells, in order to forecast the production of liquids, namely water, gas, or oil, from the producer wells. Subsequently, the output of the LSTM layer undergoes processing through a Fully Connected (FC) layer, generating 60 anticipated values corresponding to oil, gas, and water production. Following this, a reshape layer is engaged to modify the output configuration to align with the structure of the Integrated Production Model (IPM) guidance path. The attention weights derived from the IPM guidance path are then applied to the output of this trajectory. This trajectory is implemented for the model's training and testing phases.

Fig. 2 depicts the proposed model. The first step is to compute the IPMs as input to the model from the first path. The injection ratio is provided to the LSTM model to predict the production rate in the second path. In particular, the white circle refers to the multiplication process in the element.

4. Experimental results and discussion

We evaluated the proposed model using time steps of injected water or CO2 and produced oil, water, and gas collected from the reservoir. TensorFlow is used to evaluate our model-based connectivity estimations. The rectified linear unit (ReLU) activation function was used in the convolution layers, and then the softmax layer was added for normalisation to obtain the attention weights; the number of epochs was set to 100 and Adam optimiser was used. 80% of the dataset is used to train the model, in which we compute the IPM for the training phase. 20% of the data set is used to test and evaluate the model, in which we only use the injected liquid ratio and do not need to compute IPM in this phase.

Table 2
Obtained results from the proposed method.

Metrics	Injecting liquid	Oil (±SD)	Water (±SD)	Gas (±SD)
Acc. (%)	CO2	80.1 (±1.5)	79.9 (±1.2)	82.1 (±1.3)
	WAG	82.1 (±1.0)	86.8 (±1.1)	86.9 (±1.2)
	Water	75.9 (±1.4)	78.2 (±1.3)	77.4 (±1.5)
MSE	CO2	0.0035 (±0.0002)	0.007 (±0.0003)	0.0012 (±0.0001)
	WAG	0.0023 (±0.0001)	0.003 (±0.0002)	0.0001 (±0.00005)
	Water	0.15 (±0.01)	0.027 (±0.003)	0.0025 (±0.0002)

4.1. Results

The findings obtained by our methodology are described in Table 2. We systematically investigated three scenarios concerning using CO2 as an injector liquid. In this context, the accuracy rates for the produced oil, water, and gas were 80.1%, 79.9%, and 82.1%, respectively. The corresponding mean square error (MSE) values were estimated at 0.0035, 0.007, and 0.0012, respectively.

Proceeding to the second scenario, we examined the model's performance when water was the injected liquid. In this instance, the accuracy rates for the produced oil, water, and gas were 75.9%, 78.2%, and 77.4%, respectively. The concurrent MSE values for these sorts were observed to be 0.15, 0.027, and 0.0025, respectively.

Table 2 presents these quantitative results that indicate the model's ability to operate under irregular injector liquid conditions. The accuracy in predicting oil, water and gas production, along with the associated MSE values, presents a valuable understanding of the efficacy of the proposed method. Further investigation and discussion will explore the subtle importance of these results within the broader context of reservoir management and hydrocarbon production.

In the third scenario, where the water-alternating gas (WAG) technique was employed, our investigation demonstrated notable accuracy rates for the produced components. The accuracy for oil production was calculated at 82.1%, water production at 86.8%, and gas production at 86.9%. As illustrated in Table 2, these values represent the model's ability to forecast outputs when using the WAG injection procedure accurately. The MSE values associated with WAG injection were also of concern. The MSE for oil production was estimated at 0.0023, water production at 0.003, and gas production at 0.0001. These MSE values provide an understanding of the exactness of the model predictions under the dynamic requirements of WAG injection, further contributing to our understanding of the effectiveness of the suggested methodology in reservoir management and optimisation of hydrocarbon production.

Our qualitative research aimed to align oil, water and gas production outcomes using water and gas to inject fluids. Fig. 3 corresponds to the quantities of oil, water and gas made and the corresponding predicted values. This alignment is indicated across two sequential time steps regarding the injection of water and CO2 and confines data from 20 producer wells. The illustrated presentation in Fig. 3 delivers an understanding of the concordance between the predicted and actual production finds for each component, explaining the significance of employing water and CO2 as injecting fluids in the reservoir structure.

4.2. CRM-based AGFM assessment

The Capacitance Resistance Model (CRM) (Albertoni and Lake, 2003) allows evaluation of rapid water flood and CO2 flooding performance. This semianalytical modelling approach is based on signal processing and utilises a generalised non-linear multivariate regression approach. A difference in the injection rate in a well signals a signal, and the following reaction is monitored in one or more production wells. If available, the CRM employs data on production, injection rate, and bottom-hole pressure to fine-tune the model for a respective reservoir. Tracking calibration, the model is then utilised to make predictions.

CRM considers inter-well connectivity in a (CO2) Water Alternating Gas (WAG). WAG is a reservoir control strategy typically employed in enhanced oil recovery (EOR) and improved oil recovery (IOR) approaches. The WAG method involves alternating injections of water and gas into an oil reservoir. This cyclic injection way is prepared to enhance oil recovery efficiency compared to conventional methods. Thus, we evaluate our model without separating the time steps of flooding water or CO2. CRM equation is expressed as in Eq. (7).

$$q(t_n) = q(t_{n-1})e^{-\Delta t/\tau + 1 - e^{-\Delta t/\tau}} \times \prod_{i=1}^{n} f_i \times i \times j_i \tag{7}$$

where $q(t_n)$ is the predicted production rate at time t_n, $q(t_{n-1})$ is the actual production rate at the previous time, Δt is the time step, f_i is a coefficient associated with the ith injector, and inj_i is the injection rate of ith injector.

The alignment of the oil, water, and gas produced was thoroughly compared with the fluids extracted and the predictions derived from the CRM and our AFGM. Figs. 4–6 encapsulate the differences between the fluids produced through CRM and AFGM compared to the factual quantities extracted over 20 consecutive time steps. These visual representations aggregate the accumulative outputs from all production wells. Remarkably, our proposed AFGM indicates a superior concordance with the actual production, surpassing the predictions derived from CRM.

4.3. Inter-well connectivity

Interwell connectivity refers to the degree of impact or association between different wells in a reservoir. Comprehending and representing inter-well connectivity is essential in reservoir management and optimisation procedures. These connections allow us to manage strategies to enhance oil recovery (EOR) to improve production rates.

The graph in Fig. 7 illustrating associations between injectors and producers, with line thickness approximating the connection's strength, offers a visual presentation of inter-well connectivity within a reservoir. This visualisation is a valuable tool in reservoir engineering and control. Operators can enhance fluid displacement efficiency and overall reservoir performance by placing injectors and producers in areas with robust connectivity. Strong relationships enable enhanced reservoir sweep efficiency. Fluids injected into the reservoir can effectively displace hydrocarbons toward production wells, guiding a more extensive recovery of oil and gas resources.

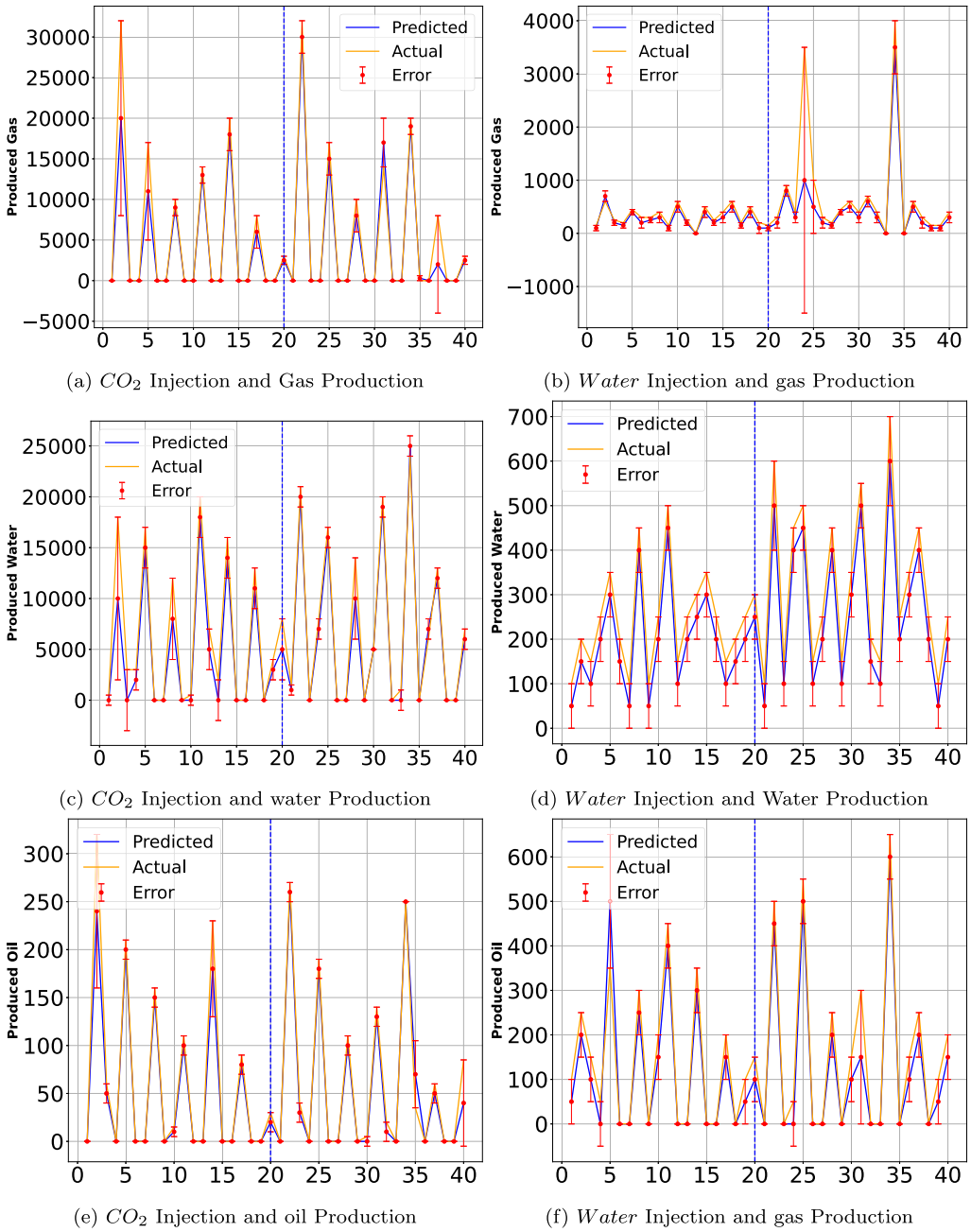
The connection plot allows us to identify areas where channelling effects occur. Channelling is the preferential flow of fluids injected along specific paths. Operators can mitigate channelling and promote uniform fluid distribution by comprehending and handling solid connections.

To see the impact of changing some injector wells to producers, we changed 2,4 and 6 to act as producers, and producers numbered 3,5 and 15 to act as injectors. The proposed model suggests more impact on inter-well connectivity, and more connected wells are presented in Fig. 8.

4.4. Comparison with state-of-the-art

In this section, we evaluate the state-of-the-art method based on the well topology available in this article and compare it with the proposed method based on the WAG scenario, in which we inject water and CO2 alternatively (see Table 3).

The GSH-BOS method accurately predicts oil, water and gas production, with values 74.2%, 69%, and 0.012, respectively. The MSE values for oil, water and gas production are 0.05, 0.09, and 0.012, respectively. Although GSH-BOS has appropriate accuracy, its MSE values indicate a specific deviation from the actual values, particularly for



(a) CO_2 Injection and Gas Production

(b) $Water$ Injection and gas Production

(c) CO_2 Injection and water Production

(d) $Water$ Injection and Water Production

(e) CO_2 Injection and oil Production

(f) $Water$ Injection and gas Production

Fig. 3. Matching the actual with predicted results (left): produced components when CO2 is used as injection fluid (right) produced components when water is used.

water and gas production. The LSTM model provides higher precision in predicting oil, water, and gas production with values of 79.8%, 80. 3% and 74%, respectively. However, the interrelated MSE values for oil (0.04), water (0.06), and gas (0.047) indicate a higher level of error corresponding to the proposed AGFM method. The LSTM model performs reasonably well, but it demonstrates challenges in accurately predicting gas production. The ANN method yields precision values of 80.1%, 81%, and 77% for oil, water and gas production. However, the MSE values for oil (0.078), water (0.09) and gas (0.054) are comparatively higher than those of the suggested AGFM method. This indicates that while ANN delivers appropriate accuracy, its predictive precision, particularly for oil and water production, is exceeded by the proposed AGFM approach. The RNN model performs accuracy percentages of 78.4%, 79%, and 80% for oil, water, and gas production, respectively. The MSE values for oil (0.065), water (0.045), and gas (0.074) reveal an intermediate level of prediction error. RNN works well in terms of accuracy but demonstrates challenges in accurately predicting water production. The suggested AGFM method exceeds the state-of-the-art methods in all metrics. It performs with the highest precision for the production of oil (82. 1%), water (86. 8%), and gas (86. 9%). Furthermore, the MSE values for oil (0.0023), water (0.003) and gas (0.0001) are substantially lower compared to the other methods. This demonstrates the outstanding predictive accuracy of the proposed AGFM method in catching the complex interplay of factors that affect oil, water, and gas production in the reservoir.

This robustness of the AGFM model can be attributed to its ability to effectively capture complex spatial and temporal dependencies between the injector and the producer wells utilising its two-path architecture. Incorporating IPM guidance in the first path permits the model to concentrate on essential inter-well exchanges often ignored by traditional models, such as LSTM and RNN, which depend heavily on sequential data without spatial context. The attention mechanism

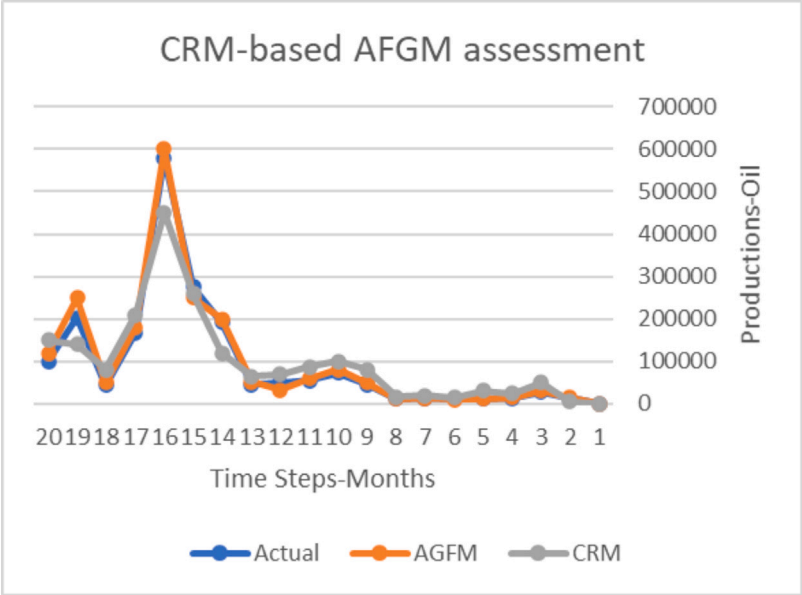


Fig. 4. CRM-based AFGM evaluation for oil production.

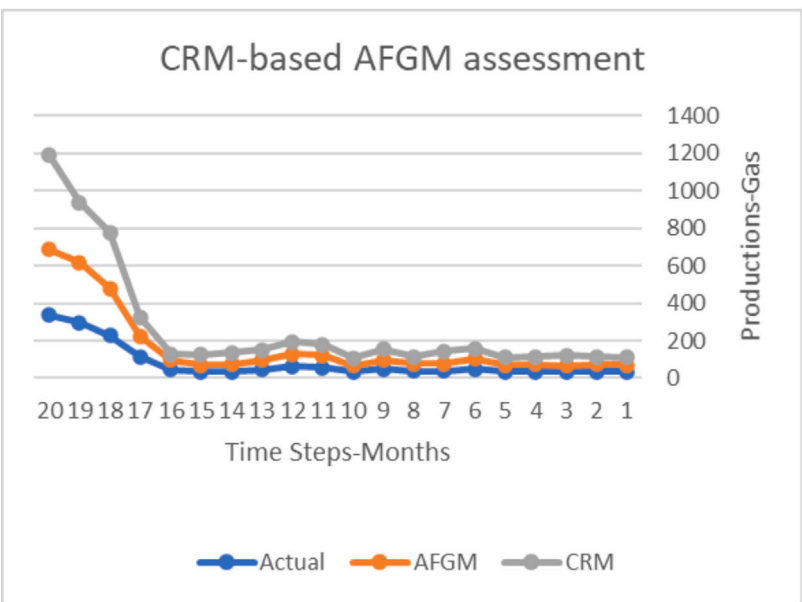


Fig. 5. CRM-based AFGM evaluation for gas production.

also refines this emphasis, improving the model's capability to highlight the most effective relationships. Moreover, the LSTM-based second path leverages these attention weights to accurately forecast production directions by preserving long-term dependencies, resulting in lower MSE values than traditional models. This spatial and temporal feature learning mix allows AGFM to reach higher prediction accuracy. The shallow MSE values for the AGFM (e.g. 0.0001 for gas production) also emphasise its precision, making it an ideal solution for predicting production in complicated reservoir settings.

The AGFM method clearly improves over existing approaches in accurately forecasting inter-well connectivity metrics. Its higher accuracy and significantly lower MSE values highlight its efficacy in optimising well-placement procedures for enhanced hydrocarbon recovery. The proposed AGFM method reveals a promising possibility for practical applications in reservoir engineering, providing a more accurate understanding of production output and contributing to refining reservoir management practices.

Accurately calculating inter-well connectivity allows appropriate injector placement and fluid allocation plans, leading to more efficient reservoir control and enhanced oil recovery. In addition, accurate production forecasting will enable operators to make proactive decisions, such as adapting injection rates and determining potential issues earlier, lowering operational costs, and optimising reservoir performance. However, the proposed model relies on clean and well-labelled data, which may restrict real-time performance in large-scale reservoirs.

4.5. Statistical test

The study comprehensively evaluated the method presented by approximating it with two primary methods, LSTM and ANN, which are considered notable in the current state of the art. To confirm the fairness of the comparison, LSTM and ANN were trained using the same

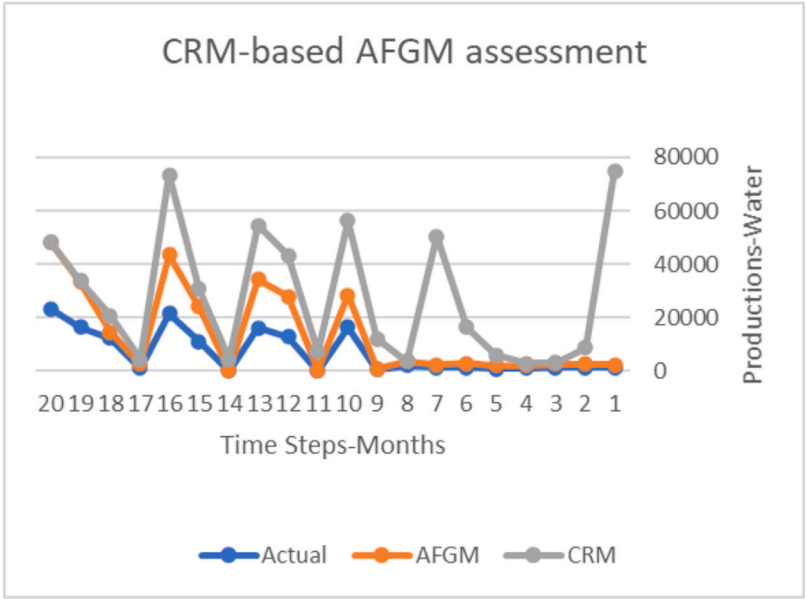


Fig. 6. CRM-based AFGM evaluation for water production.

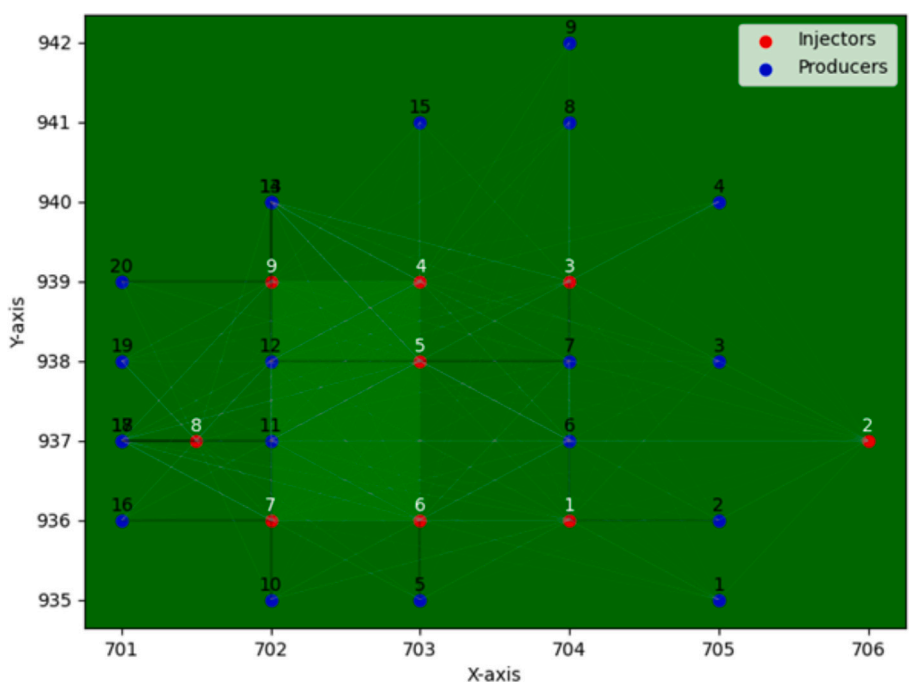


Fig. 7. Location-based injectors and producers association (red circles): injector wells (blue circles): producer wells.

data set used for AGFM, and the mean accuracy for the three types of fluids (oil, gas, and water) was assessed in the evaluation.

Each method experienced ten runs during the testing step to achieve a statistically powerful analysis, and the impacts were meticulously recorded for subsequent statistical analysis. Analysis of variance (ANOVA), a statistical method used to assess differences among group means in a sample, was used for this purpose, which is specifically applicable when comparing means across multiple groups simultaneously.

The null hypothesis, claiming the equality of means for the three models (LSTM, ANN, AGFM), was juxtaposed against the alternative hypothesis, positing that at least one means varies. The resulting low p-value (0.00139) was directed toward rejecting the null hypothesis. This proves the AGFM model demonstrates a significantly higher mean score than LSTM and ANN. Marked differences are unlikely to be attributed to random chance alone.

Fig. 9 presents box plots that visually represent the distribution of scores for AGFM, LSTM, and ANN. The box plot allows us to observe the spread of the scores, place notable differences in the central direction, and consider the variability of the model. In particular, the AGFM model has the highest median and a reasonably narrow interquartile range (IQR), demonstrating greater consistency of scores. In contrast, LSTM and ANN demonstrate some variability, as indicated by wider IQRs and the sight of an outlier for each. The ANN model displays the most comprehensive score spread and an outlier with a score of 89. This graphical examination, complemented by statistical rigour, highlights the favourable performance characteristics of the AGFM

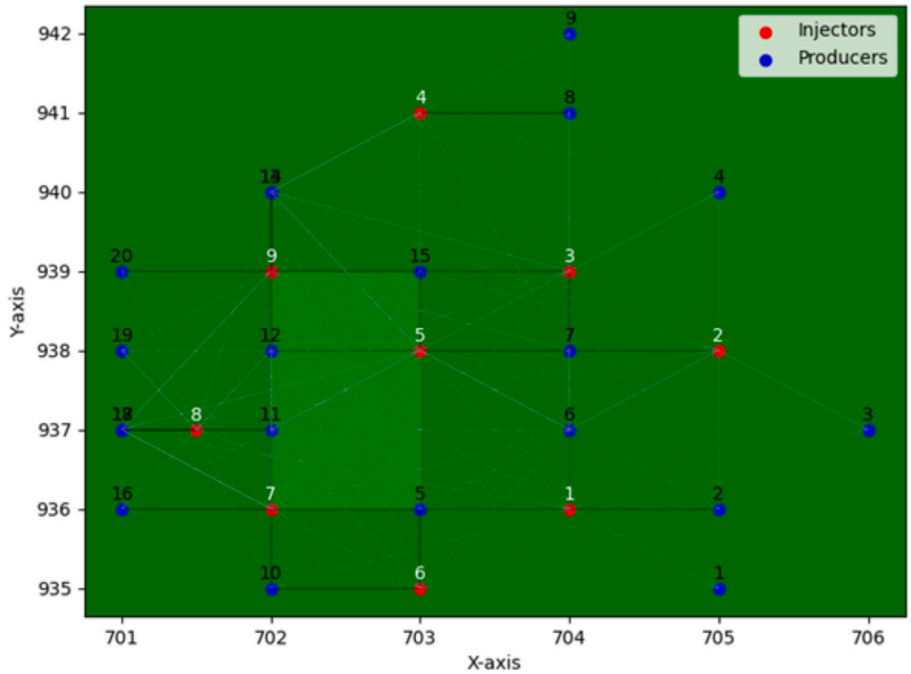


Fig. 8. Enhanced inter-well connectivity based on the proposed AGFM model suggestion.

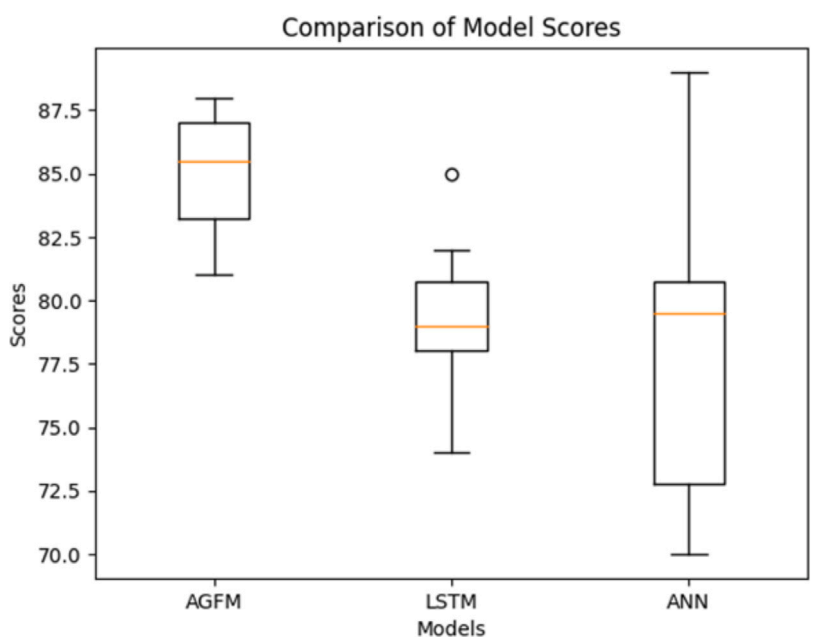


Fig. 9. The box plots for comparison of model scores.

model compared to LSTM and ANN.

5. Conclusion

The extensive evaluation of the proposed AGFM model against selected counterparts, LSTM and ANN, has generated insightful conclusions. Through rigorous statistical investigations, including the application of ANOVA, we sought to determine slight dissimilarities in performance between multiple runs. Both statistical research and visual inspections converge to declare the promising characteristics of the AGFM model in capturing complex relationships within reservoir datasets. Solid statistical proof and the visual depiction of score distributions substantiate AGFM's efficacy in optimising inter-well connectivity and enhancing hydrocarbon recovery. These conclusions contribute to the growing body of knowledge in reservoir engineering, highlighting the possibility of AGFM as a good tool for reservoir simulation and management. Future research attempts may further investigate the adaptability and scalability of AGFM in various reservoir requirements and layouts. Evaluating the accuracy and MSE metrics in different models, including GSH-BOS, LSTM, ANN, RNN, and our proposed

Table 3
Comparison with state-of-the-art oil, water, and gas production.

Method	Year	Accuracy (%)	MSE
Production oil			
GSH-BOS (Du et al., 2020)	2017	72.01	0.012
LSTM (Zhou et al., 2021)	2020	79.8	0.04
ANN (Du et al., 2023)	2021	80.1	0.078
RNN (Demiryurek et al., 2008)	2023	78.4	0.065
AGFM (ours)	–	82.1	0.0023
Production water			
GSH-BOS (Du et al., 2020)	2017	74.2	0.05
LSTM (Zhou et al., 2021)	2020	80.3	0.06
ANN (Du et al., 2023)	2021	81	0.09
RNN (Demiryurek et al., 2008)	2023	79	0.045
AGFM (ours)	–	86.8	0.003
Production gas			
GSH-BOS (Du et al., 2020)	2017	69	0.09
LSTM (Zhou et al., 2021)	2020	74	0.047
ANN (Du et al., 2023)	2021	77	0.054
RNN (Demiryurek et al., 2008)	2023	80	0.074
AGFM (ours)	–	86.9	0.0001

AGFM, provides invaluable insight into their respective interpretations in the prediction of hydrocarbon production. AGFM significantly improves accuracy and MSE in all types of production, exceeding the models compared. These findings underscore the effectiveness of AGFM in accurately predicting hydrocarbon production with minimal error. The robust performance of AGFM has promising implications for its practical application in reservoir engineering, contributing to improved reservoir management and optimisation strategies. In Fig. 8, injector wells (red points) and producer wells (blue points) play crucial roles in reservoir management and hydrocarbon recovery. Connectivity between the injector and producer wells is vital to reservoir performance. The connections between injector wells and producer wells represent the paths through which fluids (such as water or gas) travel within the reservoir. Inter-well connectivity measures how effectively the injected fluids from the injector wells reach and influence the production at the producer wells. The inter-well connectivity is affected by factors like (1) The distribution injector and producers reflect the spatial distribution of wells in the reservoir. Heterogeneous reservoirs, where the properties vary across space, can lead to variations in connectivity. Some connections may be more efficient than others due to geological heterogeneity. (2) The distance between the injector and producer wells can affect the connectivity. Generally, shorter distances may result in more effective fluid communication. Longer distances may require higher injection pressures, leading to increased fluid dispersion and potentially impacting the overall sweep efficiency. (3) Sweep efficiency measures how uniformly and effectively the injected fluids displace hydrocarbons toward the producing wells. Efficient connectivity ensures that injected fluids sweep through a significant portion of the reservoir, displacing oil, and enhancing recovery.

In Fig. 8, changing injector numbers 2,4 and 6 to producers and injector numbers 3,5 and 15 would improve interwell connectivity, as shown in Fig. 7.

The AGFM model performed better than existing techniques, reaching the highest accuracy and the lowest MSE values in all production scenarios. Its significance in capturing complex spatial and temporal dependencies through IPM-based guidance and attention mechanisms makes it a powerful tool for optimising production processes and improving reservoir performance.

Future works in inter-well connectivity for reservoirs can concentrate on improving prediction models by incorporating progressive machine learning and deep learning approaches, such as Graph Convolutional Networks (GCNs) and attention-based mechanisms, on catching complex spatiotemporal relationships between wells. In addition, growing hybrid models that integrate physical reservoir simulations with data-driven techniques can assist in overcoming the constraints of purely data-driven methods and enhance prediction robustness.

CRediT authorship contribution statement

Ahmed Saihood: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Tariq Saihood:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Sabah Abdulazeez Jebur:** Investigation, Validation, Writing – original draft, Writing – review & editing. **Christine Ehlig-Economides:** Conceptualization, Methodology, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Laith Alzubaidi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Yuantong Gu:** Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

Akkem, Yaganteeswarudu, Biswas, Saroj Kumar, Varanasi, Aruna, 2023a. Smart farming using artificial intelligence: A review. Eng. Appl. Artif. Intell. 120, 105899. http://dx.doi.org/10.1016/j.engappai.2023.105899.

Akkem, Yaganteeswarudu, Biswas, Saroj Kumar, Varanasi, Aruna, 2024. A comprehensive review of synthetic data generation in smart farming by using variational autoencoder and generative adversarial network. Eng. Appl. Artif. Intell. 131, 107881. http://dx.doi.org/10.1016/j.engappai.2024.107881.

Akkem, Yaganteeswarudu, Kumar, Biswas Saroj, Varanasi, Aruna, 2023b. Streamlit application for advanced ensemble learning methods in crop recommendation systems – A review and implementation. Indian J. Sci. Technol. 16 (48), 4688–4702. http://dx.doi.org/10.17485/ijst/v16i48.2850.

Al-Mudhafar, W.J., Wood, D.A., Al-Obaidi, D.A., Wojtanowicz, A.K., 2023. Well placement optimization through the triple-completion gas and downhole water sink-assisted gravity drainage (TC-GDWS-AGD) EOR process. Energies 16 (4), http://dx.doi.org/10.3390/en16041790.

Albertoni, A., Lake, L.W., 2003. Inferring interwell connectivity only from well-rate fluctuations in waterfloods. SPE Reserv. Eval. Eng. 6 (01), 6–16. http://dx.doi.org/10.2118/83381-PA.

Annan Boah, E., Kwami Senyo Kondo, O., Aidoo Borsah, A., Brantson, E.T., 2019. Critical evaluation of infill well placement and optimization of well spacing using the particle swarm algorithm. J. Pet. Explor. Prod. Technol. 9 (4), 3113–3133. http://dx.doi.org/10.1007/s13202-019-0710-1.

Carmichael, J.P., Liao, Y., 2022. Application of deep neural networks to distribution system state estimation and forecasting. Front. Sustain. Cities 3 (January), 1–14. http://dx.doi.org/10.3389/frsc.2021.814037.

Chen, G., Luo, X., Jiao, J.J., Xue, X., 2022. Data-driven evolutionary algorithm for oil reservoir well-placement and control optimization. Fuel 326.

Cheng, H., He, Y., Zeng, P., Li, S., Vyatkin, V., 2023. Deep learning-based prediction of subsurface oil reservoir pressure using spatio-temporal data. In: IECON 2023-49th Annual Conference of the IEEE Industrial Electronics Society. pp. 1–6. http://dx.doi.org/10.1109/IECON51785.2023.10312480.

Cheng, H., Vyatkin, V., Osipov, E., Zeng, P., Yu, H., 2020. LSTM based EFAST global sensitivity analysis for interwell connectivity evaluation using injection and production fluctuation data. IEEE Access 8, 67289–67299. http://dx.doi.org/10.1109/ACCESS.2020.2985230.

Demiryurek, U., Banaei-Kashani, F., Shahabi, C., Wilkinson, F., 2008. Neural-network based sensitivity analysis for injector-producer relationship identification. In: Intelligent Energy Conference and Exhibition. pp. SPE–112124–MS. http://dx.doi.org/10.2118/112124-MS.

Du, S., et al., 2020. The connectivity evaluation among wells in reservoir utilizing machine learning methods. IEEE Access 8, 47209–47219. http://dx.doi.org/10.1109/ACCESS.2020.2976910.

Du, S.-Y., et al., 2023. Data-driven production optimization using particle swarm algorithm based on the ensemble-learning proxy model. Pet. Sci. http://dx.doi.org/10.1016/j.petsci.2023.04.001.

Framework, L., 2021. Fast optimization of injector selection for waterflood, CO 2 -EOR and storage using an innovative machine learning framework.

Hamidi, H., Rafati, R., 2012. Prediction of oil reservoir porosity based on BP-ANN. In: ICIMTR 2012-2012 International Conference on Innovation, Management and Technology Research. pp. 241–246. http://dx.doi.org/10.1109/ICIMTR.2012.6236396.

Hernandez-Ambato, J., Asqui-Santillan, G., Arellano, A., Cunalata, C., 2017. Multistep-ahead streamflow and reservoir level prediction using ANNs for production planning in hydroelectric stations. In: 2017 16th IEEE International Conference on Machine Learning and Applications. ICMLA, pp. 479–484. http://dx.doi.org/10.1109/ICMLA.2017.0-115.

Huang, Zhao Qin, Wang, Zhao Xu, Hu, Hui Fang, Zhang, Shi Ming, Liang, Yong Xing, Guo, Qi, Yao, Jun, 2024. Dynamic interwell connectivity analysis of multi-layer waterflooding reservoirs based on an improved graph neural network. Pet. Sci. 21 (2), 1062–1080. http://dx.doi.org/10.1016/j.petsci.2023.11.008.

Jiang, Y., Shen, W., Zhang, H., Zhang, K., Wang, J., Zhang, L., 2023a. An interpretable recurrent neural network for waterflooding reservoir flow disequilibrium analysis. Water 15 (4), http://dx.doi.org/10.3390/w15040623.

Jiang, Yunqi, Zhang, Huaqing, Zhang, Kai, Wang, Jian, Han, Jianfa, Cui, Shiti, Zhang, Liming, Zhao, Hanjun, Liu, Piyang, Song, Honglin, 2023b. Waterflooding interwell connectivity characterization and productivity forecast with physical knowledge fusion and model structure transfer. Water (Switz.) 15 (2), http://dx.doi.org/10.3390/w15020218.

Jiang, Y., et al., 2023c. Waterflooding interwell connectivity characterization and productivity forecast with physical knowledge fusion and model structure transfer. Water 15 (2), http://dx.doi.org/10.3390/w15020218.

Kashkooli, S.B., Gandomkar, A., Riazi, M., Tavallali, M.S., 2022. Coupled optimization of carbon dioxide sequestration and CO2 enhanced oil recovery. J. Pet. Sci. Eng. 208 (July), 109257. http://dx.doi.org/10.1016/j.petrol.2021.109257.

Ma, Y., Cao, D., Feng, J., 2023. Application of random forest algorithm with Bayesian optimization to reservoir prediction: Application of random forest algorithm with Bayesian optimization to reservoir prediction. In: Proceedings of 2023 International Conference on New Trends in Computational Intelligence. NTCI 2023, Vol. 1, pp. 412–418. http://dx.doi.org/10.1109/NTCI60157.2023.10403755.

Nwachukwu, A., Omoregbee, H.O., Okwu, M.O., Tartibu, L.K., Enarevba, R., Adesuji, A., 2022. A review of artificial neural network applications in petroleum exploration, production and distribution operations. Int. J. Sci. Technol. Res. 11 (01), 1, [Online]. Available: www.ijstr.org.

Onuh, C.Y., Alaigba, D., Rotimi, O.J., Arowolo, B.T., 2017. Well placement optimization using a basic genetic search heuristics algorithm and a black oil simulator. Open J. Yangtze Oil Gas 02 (04), 214–225. http://dx.doi.org/10.4236/ojogas.2017.24017.

Paul, T., et al., 2021. Forecasting of reservoir inflow by the combination of deep learning and conventional machine learning. In: 2021 International Conference on Data Mining Workshops. ICDMW, pp. 558–565. http://dx.doi.org/10.1109/ICDMW53433.2021.00074.

Ramalho, L.A.M., Tueros, J.A.R., Horowitz, B., 2024. Machine learning applied to evaluation of reservoir connectivity. Neural Comput. Appl. 36 (2), 731–746. http://dx.doi.org/10.1007/s00521-023-09056-0.

Saihood, Tariq, Saihood, Ahmed, Al-Shaher, Mohamed Adel, Ehlig-Economides, Christine, Zargar, Zeinab, 2024. Fast evaluation of reservoir connectivity via a new deep learning approach: Attention-based graph neural network for fusion model. In: Atce. p. 12. http://dx.doi.org/10.2118/221029-MS.

Saikia, P., Baruah, R.D., Singh, S.K., Chaudhuri, P.K., 2020. Artificial Neural Networks in the domain of reservoir characterization: A review from shallow to deep models. Comput. Geosci. 135, 104357. http://dx.doi.org/10.1016/j.cageo.2019.104357.

Sen, D., Chen, H., Datta-Gupta, A., 2022a. Inter-well connectivity detection in CO2 WAG projects using statistical recurrent unit models. Fuel 311, 1–37.

Sen, D., Chen, H., Datta-Gupta, A., 2022b. Inter-well connectivity detection in CO2 WAG projects using statistical recurrent unit models. Fuel 311, 122600. http://dx.doi.org/10.1016/j.fuel.2021.122600.

Wang, B., Li, T., Xu, N., Zhou, H., Xiong, Z., Long, W., 2021. A novel reservoir modeling method based on improved hierarchical XGBoost. In: IEEE Advanced Information Technology, Electronic and Automation Control Conference. IAEAC, Vol. 5, pp. 1918–1923. http://dx.doi.org/10.1109/IAEAC50856.2021.9390679.

Wang, D., et al., 2017. Ensemble-based optimization of interwell connectivity in heterogeneous waterflooding reservoirs. J. Nat. Gas Sci. Eng. 38, 245–256. http://dx.doi.org/10.1016/j.jngse.2016.12.030.

Xiong, X., Lee, K.J., 2020. Data-driven modeling to optimize the injection well placement for waterflooding in heterogeneous reservoirs applying artificial neural networks and reducing observation cost. Energy Explor. Exploit. 38 (6), 2413–2435. http://dx.doi.org/10.1177/0144598720927470.

Yan, W., Ruogu, W., Shengyi, Y., Jianping, L., 2021. Tight reservoirs classification using random forest: A case study of he 8 member in eastern yan'an gas field. In: 2021 3rd International Conference on Intelligent Control, Measurement and Signal Processing and Intelligent Oil Field. ICMSP 2021, pp. 288–292. http://dx.doi.org/10.1109/ICMSP53480.2021.9513413.

Yuan, J., Zeng, X., Wu, H., Zhang, W., Zhou, J., Chen, B., 2021. Analytical determination of interwell connectivity based on interwell influence. Tsinghua Sci. Technol. 26 (6), 813–820. http://dx.doi.org/10.26599/TST.2020.9010039.

Zakaria, A.S., Hafez, M.M., Ochi, J., Zaki, K.S., Loloi, M., Abou-Sayed, A., 2011. Application of genetic algorithms to the optimization of pressure transient analysis of water injectors using type curves. In: SPE European Formation Damage Conference. pp. SPE–143386–MS. http://dx.doi.org/10.2118/143386-MS.

Zhao, Y., Luo, R., Li, L., Zhang, R., Zhang, D., Zhang, T., et al., 2024. A review on optimization algorithms and surrogate models for reservoir automatic history matching. Geoenergy Sci. Eng. [Internet] 233 (2023), 212554, Available from: https://doi.org/10.1016/j.geoen.2023.212554.

Zhou, Y., Xu, Y., Rao, X., Hu, Y., Liu, D., Zhao, H., 2021. Artificial neural network-(ANN-) based proxy model for fast performances' forecast and inverse schedule design of steam-flooding reservoirs. Math. Probl. Eng. 2021, 5527259. http://dx.doi.org/10.1155/2021/5527259.